

Stat 517
Homework # 1

1. For each of the following, find the constant c so that $p(x)$ satisfies the condition of being a probability mass function (pmf) of one random variable X .
 - (a) $p(x) = c(\frac{2}{3})^x, x = 1, 2, 3, \dots$, zero elsewhere.
 - (b) $p(x) = cx, x = 1, 2, 3, 4, 5, 6$, zero elsewhere.
2. If the probability density function (pdf) of X is $f(x) = 2xe^{-x^2}, 0 < x < \infty$, zero elsewhere, determine the pdf of $Y = X^2$.
3. Let X have the pdf $f(x) = 3x^2, 0 < x < 1$, zero elsewhere.
 - (a) Compute $\mathbb{E}(X^3)$.
 - (b) Show that $Y = X^3$ has a uniform(0, 1) distribution.
 - (c) Compute $E(Y)$ and compare this result with the answer obtained in part (a).
4. Let $p(x) = (\frac{1}{2})^x, x = 1, 2, 3, \dots$, zero elsewhere, be the pmf of the random variable X . Find the moment generating function (mgf), the mean, and the variance of X .
5. Let $f(x_1, x_2) = 4x_1x_2, 0 < x_1 < 1, 0 < x_2 < 1$, zero elsewhere, be the pdf of X_1 and X_2 . Find $P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1), P(X_1 = X_2), P(X_1 < X_2)$, and $P(X_1 \leq X_2)$.
6. Let $f(x, y) = e^{-x-y}, 0 < x < \infty, 0 < y < \infty$, zero elsewhere, be the pdf of X and Y . Then if $Z = X + Y$, compute $P(X \leq 0), P(Z \leq 6)$, and more generally $P(Z \leq z)$ for $0 < z < \infty$. What is the pdf of Z ?
7. Let X_1 and X_2 have the joint pdf $f(x_1, x_2) = 15x_1^2x_2, 0 < x_1 < x_2 < 1$, zero elsewhere. Find the marginal pdfs and compute $P(X_1 + X_2 \leq 1)$.
8. Let X and Y have the joint pdf $f(x, y) = 3x, 0 < y < x < 1$, zero elsewhere. Are X and Y independent? If not, find $E(X|y)$.

1. For each of the following, find the constant c so that $p(x)$ satisfies the condition of being a probability mass function (pmf) of one random variable X .

$$p(x) = P(X=x)$$

(a) $p(x) = c\left(\frac{2}{3}\right)^x, x = 1, 2, 3, \dots$, zero elsewhere.

(b) $p(x) = cx, x = 1, 2, 3, 4, 5, 6$, zero elsewhere.

For a function to be pmf: $p(x) \geq 0$
 $\sum_x p(x) = 1$

a) $p(x) = c\left(\frac{2}{3}\right)^x, x = 1, 2, 3, \dots$

$$\sum_{x=1}^{\infty} c\left(\frac{2}{3}\right)^x = 1$$

$$c \sum_{x=1}^{\infty} \left(\frac{2}{3}\right)^x = 1$$

using geometric series $\sum_{r=1}^{\infty} r^x = \frac{r}{1-r}$

$$\sum_{x=1}^{\infty} \left(\frac{2}{3}\right)^x = \frac{2/3}{1-2/3} = \frac{2/3}{1/3} = 2$$

$$2c = 1$$

$$c = 1/2$$

b) $p(x) = CX, x = 1, \dots, 6$

$$\sum_{x=1}^6 p(x) = 1$$

$$\sum_{x=1}^6 CX = 1$$

$$C \sum_{x=1}^6 x = 1$$

$$C(1+2+3+4+5+6) = 1$$

$$21C = 1$$

$$C = 1/21$$

2. If the probability density function (pdf) of X is $f(x) = 2xe^{-x^2}, 0 < x < \infty$, zero elsewhere, determine the pdf of $Y = X^2$.

$$X = \sqrt{Y} \rightarrow X \text{ as a function of } Y$$

$$0 < \sqrt{Y} < \infty$$

$$0 < Y < \infty$$

$$\frac{dx}{dy} = \frac{d}{dy} y^{1/2} \rightarrow \text{take derivative of } X \text{ to } Y$$

$$\frac{d}{dy} y^a = ay^{a-1} \rightarrow \text{power rule}$$

$$\frac{dx}{dy} = \frac{1}{2} y^{-1/2} = \frac{1}{2\sqrt{Y}} \rightarrow a = 1/2 \text{ plug in}$$

Transformation Formula: $f_Y(x) = f_X(\sqrt{Y}) \left| \frac{dx}{dy} \right|$
 $= 2\sqrt{Y} e^{-(\sqrt{Y})^2} \left(\frac{1}{2\sqrt{Y}} \right) = e^{-Y}$

$$f_Y(Y) = \begin{cases} e^{-Y}, & Y > 0 \\ 0, & \text{otherwise} \end{cases}$$

3. Let X have the pdf $f(x) = 3x^2$, $0 < x < 1$, zero elsewhere.

(a) Compute $E(X^3)$.

(b) Show that $Y = X^3$ has a uniform(0, 1) distribution.

(c) Compute $E(Y)$ and compare this result with the answer obtained in part (a).

$$a) E(X^3) = \int_0^1 x^3 f_X(x) dx = \int_0^1 x^3 (3x^2) dx = 3 \int_0^1 x^5 dx = 3 \left[\frac{x^6}{6} \right]_0^1 = 3/6$$

$$E(X^3) = 1/2$$

$$b) F_Y(y) = P(Y \leq y) \\ = P(X^3 \leq y) \\ = P(X \leq y^{1/3})$$

$$\text{For } 0 < y < 1, F_Y(y) = \int_0^{y^{1/3}} 3x^2 dx \\ = [x^3]_0^{y^{1/3}} = y$$

$$F_Y(y) = y, 0 < y < 1 \\ Y \sim \text{Uniform}(0, 1)$$

$$c) Y \sim \text{Uniform}(0, 1)$$

$$E(Y) = \frac{0+1}{2}$$

$$E(Y) = 1/2 = E(X^3)$$

4. Let $p(x) = (\frac{1}{2})^x$, $x = 1, 2, 3, \dots$, zero elsewhere, be the pmf of the random variable X . Find the moment generating function (mgf), the mean, and the variance of X .

$$\text{mgf: } M(t) = E(e^{tx})$$

$$= \sum_{x=1}^{\infty} e^{tx} \left(\frac{1}{2}\right)^x = \sum_{x=1}^{\infty} \left(\frac{e^t}{2}\right)^x$$

$$\text{As a geometric series: } M(t) = \frac{e^t/2}{1 - e^t/2}$$

$$M(t) = \frac{e^t}{2 - e^t}, t < \ln 2$$

$$\text{mean: } E(X) = M'(0) \\ M(t) = e^t / (2 - e^t)$$

$$\text{Differentiate: } M'(t) = \frac{e^t(2 - e^t) - e^t(-e^t)}{(2 - e^t)^2} = \frac{2e^t}{(2 - e^t)^2}$$

$$\text{At } t=0: M'(0) = \frac{2}{(2-1)^2} = 2$$

$$E(X) = 2$$

$$\text{variance: } * \text{Var}(X) = E(X^2) - [E(X)]^2 \text{ and } E(X^2) = M''(0)$$

$$\text{Differentiate: } M'(t) = \frac{e^t(2 - e^t) - e^t(-e^t)}{(2 - e^t)^2}$$

$$\text{At } t \neq 0: M''(0) = \frac{2}{1^2} + \frac{4}{1^3} = 6$$

$$* \text{Var}(X) = 6 - (2)^2$$

$$\text{Var}(X) = 2$$

$$M''(t) = \frac{2e^t}{(2 - e^t)^2} + \frac{4e^{2t}}{(2 - e^t)^3}$$

5. Let $f(x_1, x_2) = 4x_1x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$, zero elsewhere, be the pdf of X_1 and X_2 .

Find $P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1)$, $P(X_1 = X_2)$, $P(X_1 < X_2)$, and $P(X_1 \leq X_2)$.

a)

b)

c)

d)

$$a) P(0 < X_1 < \frac{1}{2}, \frac{1}{4} < X_2 < 1)$$

$$= \int_0^{\frac{1}{2}} \int_{\frac{1}{4}}^1 4x_1x_2 dx_2 dx_1 = \int_0^{\frac{1}{2}} 4x_1 \left[\frac{x_2^2}{2} \right]_{\frac{1}{4}}^1 dx_1 = \int_0^{\frac{1}{2}} 2x_1 (1 - \frac{1}{16}) dx_1 = \int_0^{\frac{1}{2}} 2x_1 (\frac{15}{16}) dx_1$$

$$= \int_0^{\frac{1}{2}} \frac{15}{8} x_1 dx_1 = \frac{15}{8} \left[\frac{x_1^2}{2} \right]_0^{\frac{1}{2}} = \frac{15}{8} \cdot \frac{1}{8} = \frac{15}{64}$$

$$b) P(X_1 = X_2)$$

X_1 and X_2 are continuous, therefore: $P(X_1 = X_2) = 0$

$$c) P(X_1 < X_2) = \int_0^1 \int_0^{x_2} 4x_1x_2 dx_1 dx_2 = \int_0^1 4x_2 \left[\frac{x_1^2}{2} \right]_0^{x_2} dx_2 = \int_0^1 2x_2^3 dx_2 = 2 \left[\frac{x_2^4}{4} \right]_0^1$$

$$P(X_1 < X_2) = \frac{1}{2}$$

$$d) P(X_1 \leq X_2)$$

$$\text{Use } P(X_1 \leq X_2) = P(X_1 < X_2) + \cancel{P(X_1 = X_2)} = \frac{1}{2} + 0$$

$$P(X_1 \leq X_2) = \frac{1}{2}$$

6. Let $f(x, y) = e^{-x-y}$, $0 < x < \infty$, $0 < y < \infty$, zero elsewhere, be the pdf of X and Y . Then if $Z = X + Y$, compute $P(X \leq 0)$, $P(Z \leq 6)$, and more generally $P(Z \leq z)$ for $0 < z < \infty$. What is the pdf of Z ?

a) $P(X \leq 0) = 0$
no prob below 0

b) $P(Z \leq 6) = P(X + Y \leq 6)$
for $z > 0$, $P(Z \leq z) = \int_0^z \int_0^{z-x} e^{-x-y} dy dx$

Integrate to y :

$$= \int_0^z e^{-x} \left[\int_0^{z-x} e^{-y} dy \right] dx = \int_0^z e^{-x} (1 - e^{-(z-x)}) dx = \int_0^z (e^{-x} - e^{-z}) dx$$

$$= [-e^{-x}]_0^z - [-e^{-z}x]_0^z = (1 - e^{-z}) - ze^{-z} = 1 - e^{-z}(1+z)$$

c) $P(Z \leq z) = 1 - e^{-z}(1+z)$, $z > 0$ hence $P(Z \leq 6) = 1 - 7e^{-6}$

d) PDF of Z

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$F_Z(z) = 1 - e^{-z}(1+z)$$

Differentiate: $f_Z(z) = ze^{-z}$

$$f_Z(z) = \begin{cases} ze^{-z}, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

7. Let X_1 and X_2 have the joint pdf $f(x_1, x_2) = 15x_1^2x_2$, $0 < x_1 < x_2 < 1$, zero elsewhere. Find the marginal pdfs and compute $P(X_1 + X_2 \leq 1)$.

a) b)

c)

a) Marginal PDF: X_1 $x_1 < x_2 < 1$
 $f_{X_1}(x_1) = \int_{x_1}^1 15x_1^2x_2 dx_2 = 15x_1^2 \left[\frac{x_2^2}{2} \right]_{x_1}^1$

$$f_{X_1}(x_1) = \frac{15}{2} x_1^2 (1 - x_1^2), \quad 0 < x_1 < 1$$

b) Marginal PDF: X_2 $0 < x_1 < x_2$
 $f_{X_2}(x_2) = \int_0^{x_2} 15x_1^2x_2 dx_1 = 15x_2 \left[\frac{x_1^3}{3} \right]_0^{x_2}$

$$f_{X_2}(x_2) = 5x_2^4, \quad 0 < x_2 < 1$$

c) $P(X_1 + X_2 \leq 1)$

$$0 < x_1 < x_2 < 1$$

ceiling 1 ceiling 2
 $0 < x_1 < x_2$, $x_1 \leq 1 - x_2$

$x_2 = 1 - x_2$
 $x_2 = 1/2$

Before and after $1/2$, smaller ceiling changes.

$$0 < x_2 < 1/2 \rightarrow \int_0^{1/2} \int_0^{x_2} 15x_1^2x_2 dx_1 dx_2$$

$$1/2 < x_2 < 1 \rightarrow \int_{1/2}^1 \int_0^{1-x_2} 15x_1^2x_2 dx_1 dx_2$$

$$P(X_1 + X_2 \leq 1) = \int_0^{1/2} \int_0^{x_2} 15x_1^2x_2 dx_1 dx_2 + \int_{1/2}^1 \int_0^{1-x_2} 15x_1^2x_2 dx_1 dx_2$$

For small x_2 , rule $x_1 < x_2$ is stricter.

For large x_2 , rule $x_1 + x_2 \leq 1$ is stricter.

$$\begin{aligned} & \int_0^{1/2} \int_0^{x_2} 15x_1^2x_2 dx_1 dx_2 \\ &= \int_0^{1/2} 15x_2 \left[\frac{x_1^3}{3} \right]_0^{x_2} dx_2 \\ &= \int_0^{1/2} 5x_2^4 dx_2 \\ &= \left[x_2^5 \right]_0^{1/2} \\ &= \frac{1}{32} \end{aligned}$$

$$\begin{aligned} & \int_{1/2}^1 \int_0^{1-x_2} 15x_1^2x_2 dx_1 dx_2 \\ &= \int_{1/2}^1 15x_2 \left[\frac{x_1^3}{3} \right]_0^{1-x_2} dx_2 \\ &= \int_{1/2}^1 5x_2(1-x_2)^3 dx_2 \\ &= \frac{3}{64} \end{aligned}$$

$$\begin{aligned} P(X_1 + X_2 \leq 1) &= \frac{1}{32} + \frac{3}{64} \\ &= \frac{2}{64} + \frac{3}{64} \\ &= \frac{5}{64} \end{aligned}$$

8. Let X and Y have the joint pdf $f(x, y) = 3x$, $0 < y < x < 1$, zero elsewhere. Are X and Y independent? If not, find $E(X|y)$. b)

a) $0 < y < x < 1$

The possible values of X depend on Y , they are not independent.

b) $E(X|y)$: Average value of X assuming $Y=y$.

$$E(X|y) = \int x \cdot f_{X|Y}(x|y) dx$$

• Fixed y at $0 < y < x < 1$, so x must be y to 1. $\int_y^1 \rightarrow f_Y(y) = \int_y^1 3x dx$

$$= 3 \int_y^1 x dx$$

$$f_Y(y) = 3 \left[\frac{x^2}{2} \right]_y^1 = \frac{3}{2}(1 - y^2) = \frac{3}{2}(1 - y^2)$$

• Conditional PDF: $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} \rightarrow \frac{3x}{\frac{3}{2}(1 - y^2)}$

$$f_{X|Y}(x|y) = \frac{3x}{\frac{3}{2}(1 - y^2)} = \frac{2x}{1 - y^2}$$

$$E(X|y) = \int_y^1 x \cdot \frac{2x}{1 - y^2} dx = \int_y^1 \frac{2x^2}{1 - y^2} dx$$

Treat y as a constant: $\frac{2}{1 - y^2} \left[\frac{x^3}{3} \right]_y^1 = \frac{2}{1 - y^2} \cdot \frac{1 - y^3}{3}$

$$E(X|y) = \frac{2(1 - y^3)}{3(1 - y^2)}$$